Neel Gali

Autonomous Vehicle Perception Engineer

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PROFESSIONAL SUMMARY

Autonomous Vehicle Perception Engineer with 3+ years of experience building software components for advanced perception modules and ML driven autonomy within autonomous vehicle stacks. Expertise in developing and deploying computer vision and machine learning techniques for 4D world representations, ensuring real improvements to driving abilities on real vehicles.

TECHNICAL SKILLS

Programming Languages: C++, Python, SQL, Bash, PowerShell

Perception & ML Frameworks: PyTorch, TensorFlow, OpenCV, YOLOv8, Detectron2, scikit-learn, Hugging Face

Autonomous Systems & Computer Vision: Perception, 4D world representations, End-to-end autonomy, Modern ML-based perception, World modeling, Object Detection, Segmentation, Reconstruction, Pre-training vision models, Diffusion Models, ML driven autonomy, Self-driving Vehicles

Cloud & DevOps: AWS, Azure, GCP, Docker, Kubernetes, Git, Jenkins, GitHub Actions, Azure DevOps, Terraform, Apache Kafka

Databases & Data Management: MySQL, PostgreSQL, MongoDB, DynamoDB, Vector Search

PROFESSIONAL EXPERIENCE

AI / ML Engineer | Epik Solutions

Jun 2025 - Present

• Architected, trained, and deployed modern ML-based perception models using Python and PyTorch to analyze high-volume sensor and telemetry event data, enabling the development of advanced 4D world representations for real-time threat detection and autonomous system monitoring; achieved 92% accuracy in identifying critical anomalies for internal security teams.

• Translated complex operational requirements into precise model objectives, evaluation metrics, and robust deployment strategies, collaborating with security and infrastructure stakeholders to integrate new perception capabilities into existing autonomy stack components.

• Prototyped and built production-ready computer vision and predictive machine learning techniques using TensorFlow and OpenCV to process diverse sensor data streams, enhancing system understanding and predictive capabilities for cybersecurity applications, directly impacting threat intelligence operations.

• Streamlined data collection to production and deployment pipelines through rigorous data cleaning, transformation, and feature extraction, ensuring high-quality inputs for downstream ML tasks and improving model training efficiency by 30%.

• Applied time-series aggregation and anomaly detection logic to identify emerging patterns across complex security telemetry, contributing to the foundational understanding required for building robust world models in dynamic environments.

• Developed automation scripts using Bash and PowerShell to support CLI-based deployments, monitoring, and operational troubleshooting for ML components, ensuring reliable functioning of perception modules within production workflows.

AI / ML Engineer | Samay Systems LLC

July 2024 - Jun 2025

• Designed and trained supervised and unsupervised ML models focused on compliance and risk analysis, using Python to construct accurate 4D world representations of system states and operational data for automated auditing; improved detection of compliance deviations by 45%.

• Developed Python services to collect, validate, and summarize evidence data from diverse APIs (AWS, Azure, GitHub, Jira), creating consistent inputs for downstream machine learning techniques and enhancing data collection to production and deployment readiness.

• Built AI-assisted workflows leveraging LLM reasoning and structured enterprise data, integrating tool-calling

agents to create sophisticated ML driven autonomy solutions for compliance validation; reduced manual review effort by 60%.

• Implemented RAG-style querying over historical compliance data utilizing vector databases, enabling natural-language interaction and contextual answers, which supported a more intuitive approach to understanding complex world models and data relationships.

• Designed and optimized complex relational schemas and SQL workflows to support multi-tenant, multi-framework compliance validation at scale, ensuring robust data foundations for perception and analytical tasks.

• Deployed ML services using AWS cloud-native infrastructure, Docker, Kubernetes, and CI/CD pipelines, ensuring high reliability and scalability across multiple production environments for critical autonomy stack components.

ML Intern | Monarch Tractor

Jun 2023 - Aug 2023

• Supported the development of advanced computer vision and ML pipelines for semi-autonomous agricultural vehicles operating in real-world environments, directly contributing to perception capabilities for end-to-end autonomy.

• Built Python-based preprocessing pipelines to transform raw sensor and image data into training-ready datasets for object detection and prediction tasks, optimizing data collection to production and deployment efficiency by 25%.

• Developed and fine-tuned object detection and segmentation models using PyTorch, TensorFlow, and OpenCV to enhance perception and prediction use cases, validating outputs against operational constraints for autonomous systems on real vehicles.

• Tested modern ML-based perception models using CLI-driven workflows, gaining practical experience with operational ML systems and contributing to the robustness of the autonomy stack.

• Collaborated closely with engineers and stakeholders to refine requirements and validate model outputs against field expectations for production deployments, ensuring alignment with self-driving vehicles' functional specifications.

• Deployed perception models to cloud-hosted production environments, gaining hands-on experience with performance monitoring and inference validation for critical autonomous vehicle components, enhancing system reliability.

Associate Data Scientist | ZeMoSo Technologies

Jul 2021 - Aug 2022

• Contributed to a cloud-based scientific operations platform, developing backend services and ML-adjacent pipelines that validated, enriched, and structured experimental data, laying foundational groundwork for future machine learning techniques.

• Supported Proof-of-Concept ML experimentation workflows by automating dataset preparation, feature generation, and experiment tracking, enhancing efficiency of data collection to production and deployment by 20%.

• Implemented CI/CD pipelines to validate models, enforce quality checks, and promote artifacts while maintaining auditability and security controls, ensuring reliable integration of perception-related models.

• Designed and optimized SQL-based data ingestion and transformation processes, improving data quality and accessibility for downstream analytical and ML tasks, crucial for building effective world models.

• Collaborated with product teams to define data requirements and analytical solutions, translating business needs into technical specifications for ML projects, directly impacting product feature development.

• Developed data visualization dashboards using Python libraries to communicate key insights from experimental data, supporting data-driven decision-making for scientific operations.

EDUCATION

Master’s in Computer Science | Oklahoma State University, Stillwater, OK

Bachelor’s in Computer Science | JNTU Hyderabad, India